Мирзаитов Тимур Вариант№13

1. $$\int (e^x+4)^3 e^x \, \mathrm{d}x = \int (e^x+4)^3 \, \mathrm{d}e^x = \int (e^x+4)^3 \, \mathrm{d}(e^x+4) = \int (e^x+4)^3 \, \mathrm{d}(e^x+4) = \frac{(e^x+4)^4}{4}$$

2. $$\int \frac{\mathrm{d}x}{1+\sqrt{2x+1}} = \left| \begin{matrix} t^2 = 2x+1 \\ 2t \, \mathrm{d}t = 2 \, dx \\ t \, \mathrm{d}t = dx \end{matrix} \right| = \int \frac{t \, \mathrm{d}t}{1+\sqrt{t^2}} = \int \frac{t \, \mathrm{d}t}{1+t} = \int (1 - \frac{1}{1+t}) \, \mathrm{d}t = t - \ln|1+t| +$$

   Вернемся к переменной x:

$$t^2 = 2x+1$$
$$t = \sqrt{2x+1}$$

$$t - \ln|1+t| + C = \sqrt{2x+1} - \ln(1+\sqrt{2x+1}) + C$$

3. $$\int \frac{1}{x\sqrt{\ln^3 x}} \, \mathrm{d}x = \int \frac{1}{\sqrt{\ln^3 x}} \, \mathrm{d}\ln x = \frac{-\ln^{-\frac{1}{2}} x}{\frac{1}{2}} + C = -\frac{2}{\sqrt{\ln x}} + C$$

4. $$\int (3x+18)2^x \, \mathrm{d}x = \left| \begin{matrix} u = 3x+18, & \mathrm{du} = 3\mathrm{d}x \\ \mathrm{d}v = 2^x \mathrm{d}x, & v = \frac{2^x}{\ln 2} \end{matrix} \right| =$$

$$(3x+18)\frac{2^x}{\ln 2} - \frac{3}{\ln 2}\int 2^x \mathrm{d}x = (3x+18)\frac{2^x}{\ln 2} - \frac{3 \cdot 2^x}{\ln^2 2} =$$

5. \int\frac{-13x-32}{(x+2)(x-1)(x+4)}\mathrm{d}x = \\ \begin{array}{l}

\end{array}

A = B \
C = D

6)

s

7)

s

8)

s

9)

s

10)

s